Rishi Navroop

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Education

B.E. in Computer Science and Engineering, JSSSTU 2023 – 2027
CGPA: 8.64 Mysuru, India

Intermediate (PU), Marimallappa's PU College, Mysuru – 95.83% 2023

Matriculation, Marimallappa's High School, Mysuru – 90.88% 2021

Projects

Weather Safety Assistant – Agentic LLM Advisory System GitHub | Live

Python, LangGraph, FastAPI, Pydantic, React, Vite, Open-Meteo API, OpenAI-compatible LLMs, pytest, Render

- Built a 9-node LangGraph workflow with per-session memory that turns a natural-language question into live Open-Meteo weather facts and cites the written safety policy behind every answer.
- Kept safety decisions out of the LLM: 12 versioned YAML SOPs are evaluated by a deterministic policy engine using three-valued (Kleene) logic, so missing data resolves to UNKNOWN, never a false "safe".
- Blocked hallucinations with a verification layer that checks every number and citation against fetched weather values and policy thresholds, else serves a deterministic fallback answer.
- Limited the LLM to intent extraction and wording (2 calls per turn) so injected text cannot change policy; shipped 412 deterministic tests plus 26 real-LLM evals, including prompt-injection attacks.

SpeechLens – Contrastive Speech Analytics (Multimodal AI Hackathon 2026, Track C) GitHub | Live

Python, wav2vec2 (CTC), ONNX Runtime, Praat/Parselmouth, librosa, NumPy, SciPy, FastAPI, Docker, Azure Container Apps

- Built a system that compares a speaker with a reference reading of the same text word by word and localises 11 delivery flaws (pace, pauses, pitch, fillers, skipped words, restarts) to a median 150 ms, each with measured evidence and a fix.
- Implemented wav2vec2 CTC forced alignment (exact Viterbi on CPU) and speaker-normalised prosody features; deviations are scored as robust z-scores against the reference speaker's own variability, with cross-validated gates.
- Created a 386-clip labelled dataset with a PSOLA flaw-injection engine; cross-validated F1 0.66 (precision 0.82), F1 0.85 on 20 unseen speakers, 0.03 false findings/min on clean audio.
- Verified explanations causally: repairing only the named cause removes 91% of findings ("Hear it fixed"); deployed with Docker on Azure Container Apps via GitHub Actions.

AI-Powered Student Performance Prediction & Intelligent Academic Assistant GitHub

Python, scikit-learn, XGBoost, CatBoost, Pandas, Hugging Face Transformers (FLAN-T5), Flask, React, GitHub Actions

- Built an end-to-end ML pipeline (feature engineering, imputation, one-hot encoding, scaling in a scikit-learn Pipeline) to predict student math scores from 1,000 records.
- Compared 10 regressors (Ridge, Linear, SVR, Random Forest, XGBoost, CatBoost, Gradient Boosting, AdaBoost, Decision Tree, KNN) with GridSearchCV and 5-fold CV; Ridge selected with test R^2 0.88 and MAE 4.2 marks.
- Built a model-grounded NLP assistant: what-if scenarios run through the trained model, and FLAN-T5 turns only the verified gains into explanations and study recommendations, with a template fallback.
- Served via a Flask REST API with a React UI at ∼10 ms p50 / 15 ms p95 prediction latency; GitHub Actions CI trains, tests and benchmarks on every push.

Experience

Treasurer, VISION 11/2024 – Present
JSSSTU Mysuru, India

- Managed budgeting and financial planning for student-led technical events, coordinating with organising teams.
- Led financial planning and execution for student initiatives involving 100+ participants.

Skills

Languages: Python, Java, C
AI & ML: scikit-learn, XGBoost, NumPy, Pandas, Hugging Face Transformers, LangGraph, LangChain, RAG, LLM evals
Speech & Audio: wav2vec2, CTC forced alignment, Praat/Parselmouth, librosa, ONNX Runtime
Backend & Data: FastAPI, Flask, Pydantic, Spring Boot, REST APIs, PostgreSQL, MongoDB, ChromaDB
Cloud & DevOps: Docker, GitHub Actions, Azure Container Apps, AWS (ECS Fargate, RDS), Render, Linux
AI-Assisted Development: Claude Code (agentic coding), prompt engineering, test-driven verification
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Spoken Languages: English (Professional), Kannada (Native), Hindi (Working)

Achievements

- Secured Second Runner-Up position in Ideaforge 1.0 Innovation Challenge, SJCE-STEP.